ABHI FALDU

LinkedIn | GitHub | Portfolio

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ABOUT ME

Digital technology student with an AI focus and hands-on experience building industrial AI systems, from anomaly detection in machine sensor data through real-time IIoT monitoring to production-ready MLOps pipelines. I am skilled in Python, PyTorch and Scikit-learn, and in deploying ML models with FastAPI, Docker and MLflow. Ready to contribute practical solutions in an AI or data science environment from day one.

SKILLS

ML & Modelling:	Scikit-learn · XGBoost · TensorFlow · PyTorch · NLP · Predictive Modelling
MLOps & Deployment:	MLflow · Docker · FastAPI · CI/CD (GitHub) · Evidently · MQTT · TFLite (Edge TPU)
Data & Visualisation:	Pandas · NumPy · Power BI · Grafana · Matplotlib · Jupyter Notebook
Programming & Tools:	Python · SQL · GitHub · Streamlit · InfluxDB
Soft Skills:	Analytical Thinking · Problem-Solving · Teamwork · Iterative Development · Initiative

PROJECTS

MLOps Pipeline for Predictive Maintenance 06/2026 – 10/2026
github.com/abhi-faldu/anomaly-detection-MLops

- Production-grade MLOps workflow for a machine-failure classifier (AI4I 2020, approx. 10,000 sensor records): training, experiment tracking and model registry with MLflow, including automatic champion promotion.
- Raised PR-AUC with XGBoost from 0.38 (baseline) to **0.83**; CI pipeline (GitHub) enforces PR-AUC ≥ 0.75 before every promotion.
- Deployed via FastAPI, data-drift monitoring with Evidently, containerized with Docker Compose; **38 tests** ($\sim$90% coverage).

Edge AI Waste Classification on Google Coral 04/2026 – 06/2026
github.com/abhi-faldu/sortitude

- Team project (3 people): MobileNetV2 transfer learning on TrashNet (6 classes), 92.9% validation accuracy; int8 quantization cut model size 3.3$\times$ (8.5 to 2.6 MB) at 89.7%.
- Compiled for the Edge TPU with all 67 ops on the accelerator: **3.09 ms** per image, 42$\times$ faster than the board CPU; live webcam classification at $\sim$30 FPS via a custom on-device web dashboard.

IIoT Simulator for Smart-Factory Monitoring 05/2026
github.com/abhi-faldu/iiot-smart-factory-sim

- Industry 4.0 simulation: 3 robot arms stream sensor data (temperature, vibration, motor current) over MQTT every second, real-time anomaly detection via sliding-window Z-score.
- Storage in InfluxDB, visualization via Grafana and a custom web dashboard with 6 services via Docker Compose.

Predictive Maintenance System for Industrial Robotic Bearings 03/2026 – 05/2026
github.com/abhi-faldu/robotic-bearing-pdm

- LSTM autoencoder (PyTorch) for anomaly detection in bearing vibration data (NASA IMS Bearing dataset): **F1-score 0.94** for early detection of bearing failures.
- FastAPI backend and Streamlit dashboard for real-time visualization of anomaly scores and maintenance recommendations, which are containerized with Docker Compose (3 services).

EDUCATION

Digital Technology and Management (DTM) from 10/2026
OTH Amberg-Weiden, Weiden Campus, Germany

Interdisciplinary program bridging digital technology, data science and management.

B.Sc. Applied Artificial Intelligence 10/2023 – 09/2026
Technische Hochschule Rosenheim, Germany

Focus areas: Machine Learning · Deep Learning · Neural Networks · Data Science · Database Systems · Software Engineering · IT Security

B.Tech Information Technology (partial) 01/2021 – 07/2022
Atmiya Institute of Technology & Science, Rajkot, India

Completed two semesters of foundational engineering studies.

EXPERIENCE

Sales Associate | Lidl GmbH & Co. KG | Weiden i.d.OPf. since 11/2025

- Reliable work in a high-volume retail environment while studying full-time, demonstrating strong time management and resilience.
- Building professional German-language skills and intercultural teamwork in a German-speaking work environment.
- Transferred from the Prien am Chiemsee store to Weiden in October 2026, alongside the change of university to OTH Amberg-Weiden.

Junior Data Analyst | MR Infoware | Rajkot, India 04/2023 – 10/2023

- Analyzed operational and business data to support data-driven decision-making through end-to-end EDA and data-preprocessing pipelines.
- Built Power BI dashboards to visualize KPIs and performance metrics for cross-functional stakeholders.
- Contributed to ML model evaluation and translated business requirements into actionable analytical insights.

LANGUAGES

English:	Fluent (C1)
German:	Intermediate (B1/B2)
Hindi / Gujarati:	Native

INTERESTS

Gaming | Hiking | Vibe Coding